

# Equal spherical droplets are exactly optimal within the spherical-fragmentation class

Candidate substantial partial answer to Conjecture 6.1 in arXiv:1004.4271

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**Status: candidate substantial partial result, likely valid.** For the three-dimensional liquid-drop ball energy, the minimum over every finite or countable mass fragmentation is always an equal fragmentation. The optimal number of droplets and every transition mass are explicit, and the equal state is strictly stable under mass-preserving perturbations away from transitions. Consequently the conjectured formula is exact over finite unions of spherical components. The unrestricted arbitrary-shape conjecture remains open.

## 1 Source question and the restricted result

Choksi and Peletier study the three-dimensional liquid-drop energy

$$e_0(m) = \inf_{|\Omega|=m} \left\{ P(\Omega) + \iint_{\Omega \times \Omega} \frac{dx \, dy}{4\pi|x-y|} \right\}.$$

For a ball of volume  $s$ , its value is

$$f(s) = as^{2/3} + bs^{5/3}, \quad a = (36\pi)^{1/3}, \quad b = \frac{2}{5} \left( \frac{3}{4\pi} \right)^{2/3}, \quad \frac{a}{b} = 10\pi. \quad (1)$$

Conjecture 6.1 of arXiv:1004.4271 proposes, among other assertions,

$$e_0(m) = \inf_{n \in \mathbb{N}} nf(m/n). \quad (2)$$

The source motivates the right side by equal balls separating to infinity. The complete source page is reproduced next.

Our theorem proves that *every* spherical fragmentation—including unequal radii and countably many formal droplets—has energy at least the right side of (2). It therefore completely resolves the mass-allocation part of the conjecture. What remains is to exclude arbitrary nonspherical components.

We also define the energy of one ball of volume  $m$ :

$$f(m) := (36\pi)^{1/3} m^{2/3} + \frac{2}{5} \left( \frac{3}{4\pi} \right)^{2/3} m^{5/3}.$$

Clearly, we have  $e_0(m) \leq f(m)$ . We conjecture the following scenario. There exists  $m^* > 0$ , such that for all  $m \leq m^*$ , there exists a global minimizer associated with  $e_0(m)$ , and it is a single ball of mass  $m$ . For  $m > m^*$ , a minimizer fails to exist. In fact, as  $m$  increases past  $m^*$ , the ball remains a local minimizer, but a minimizing sequence consisting of two balls of equal size that move away from each other has lower limiting energy. This separation is driven by the  $H^{-1}$  interaction energy, which attaches a positive penalty to any two objects at finite distance from each other. The limiting energy of such a sequence is simply the sum of the energies of two non-interacting balls, i.e.  $2f(m/2)$ . The critical  $m^*$  then is the only positive zero of  $f(m) - 2f(m/2)$ ,  $m^* \approx 22.066$ .

As  $m$  further increases above a certain  $m^{**} > m^*$ , a sequence consisting of three balls of equal size is a minimizing sequence for  $e_0(m)$ , with limiting value  $3f(m/3)$ ; and so on for higher values of  $n$ . Specifically, we conjecture the following:

**Conjecture 6.1.** *The minimizer associated with  $e_0(m)$  exists iff  $m \leq m^*$ , and it is a ball of mass  $m$ . Moreover, for all  $m > 0$ , we have*

$$e_0(m) = \inf_{n \in \mathbb{N}} n f(m/n).$$

*The infimum is achieved iff  $m \leq m^*$ .*

One might ask as to what is known about global minimizers. In our previous article [8] on the sharp-interface functionals, we prove that if a sequence (in  $\eta$ ) has bounded energy  $F_\eta$  then it must converge to a weighted sum of delta functions where all the weights  $m^i$  must have a corresponding minimizer of  $e_0(m^i)$ . One can readily check, via trial functions, that such a sequence exists. Thus for certain values of  $m$ , a minimizer of  $e_0(m)$  does exist. Unfortunately, our lower bound compactness argument gives no explicit range for the possible limiting weights  $m^i$ . One could also consider local minimizers, and in particular one can study the stability of balls. A calculation of the second variation using spherical harmonics (unpublished) indicates that the ball retains stability up to  $m_c \approx 62.83$ , well past the critical mass  $m^*$ .

Proving Conjecture [6.1] would for the first time provide some rigorous justification for why minimizers of the diblock copolymer problem have phase boundaries which resemble periodic constant mean curvature surfaces, supporting the idea that at small length scales, the perimeter (short-range) effects override the nonlocal (long-range) effects.

## 7 Analogous results in two dimensions

As in [8], we summarize the analogous results for  $d = 2$ . While we do not give all the details, we give the essential features which should enable the reader to complete the proofs. The fundamental difference between two and three dimensions is that the  $H^{-1}$ -norm is *critical* in two dimensions. As explained in [8], after rescaling with  $v = u/\eta^2$  this involves slaving  $\gamma$  to  $\eta$  via

$$\gamma = \frac{1}{|\log \eta| \eta^3},$$

## 2 The partition theorem

For  $m > 0$ , define

$$G(m) := \inf \left\{ \sum_{i \in I} f(m_i) : I \text{ finite or countable, } m_i > 0, \sum_{i \in I} m_i = m \right\}.$$

**Theorem 1** (exact spherical fragmentation). *For every  $m > 0$ ,*

$$G(m) = \min_{n \geq 1} F_n(m), \quad F_n(m) := n f(m/n). \quad (3)$$

*Every minimizing partition is finite and equal: it consists of  $n$  copies of  $m/n$  for a minimizing integer  $n$ . At a transition precisely two adjacent equal partitions minimize; otherwise the minimizing partition is unique up to permutation.*

*Proof.* First fix a maximum number  $k$  of positive pieces. The corresponding closed simplex is compact. A minimizer lies in the relative interior of a face; write  $N$  for the number of its positive coordinates. If  $N \geq 2$ , the Lagrange equations give

$$f'(m_i) = \lambda, \quad f'(s) = \frac{1}{3} s^{-1/3} (2a + 5bs).$$

Writing  $u = s^{1/3}$  shows that  $f'(s) = (2a/u + 5bu^2)/3$  first decreases and then increases. Thus a critical partition has at most two distinct masses. Moreover

$$f''(s) = \frac{2}{9} s^{-4/3} (-a + 5bs), \quad (4)$$

so the smaller mass lies on the concave branch. It cannot occur twice, since transferring mass between two such coordinates gives a negative second variation.

It remains to rule out one small mass  $x$  and  $q \geq 1$  copies of a larger mass  $y$ . Put  $y = r^3 x$  with  $r > 1$ . The stationarity equation  $f'(x) = f'(y)$  gives

$$x = \frac{2a}{5br(r+1)}. \quad (5)$$

Substitution into (4) yields the exact curvature ratio

$$\frac{f''(y)}{|f''(x)|} = \frac{2r+1}{r^3(r+2)} < 1. \quad (6)$$

Along the constraint-preserving variation which adds  $t$  to  $x$  and subtracts  $t/q$  from each copy of  $y$ , the second derivative is

$$f''(x) + \frac{1}{q} f''(y) < 0.$$

Every mixed critical point is therefore a saddle. Hence every finite minimum is equal on its positive support. Minimizing over the possible face sizes gives  $\min_{1 \leq n \leq k} F_n(m)$ .

To pass to countable partitions, observe that sufficiently small masses can always be merged. If  $x + y = s$  and  $t = x/s$ , then

$$\begin{aligned} f(x) + f(y) - f(s) &= as^{2/3} A(t) - bs^{5/3} B(t), \\ A(t) &= t^{2/3} + (1-t)^{2/3} - 1 > 0, \\ B(t) &= 1 - t^{5/3} - (1-t)^{5/3} > 0. \end{aligned}$$

The ratio  $A/B$  is continuous and positive on  $(0, 1)$ , is symmetric, and tends to infinity at both endpoints. It therefore has a positive minimum  $c$ . Whenever  $s < (a/b)c$ , the displayed difference is positive.

Given a countable partition, choose a tail whose total mass  $s$  is below this threshold. Repeatedly merge finite partial tails and pass to the limit; continuity of  $f$  gives

$$\sum_{i>N} f(m_i) \geq f\left(\sum_{i>N} m_i\right).$$

The resulting finite partition has no larger energy, so the finite result proves (3). An infinite partition cannot attain equality, because a sufficiently small tail containing two positive masses admits a strict merge. Finally,  $F_n(m) \rightarrow \infty$  as  $n \rightarrow \infty$ , so the discrete minimum is attained.  $\square$

### 3 Exact transition masses

Since

$$F_n(m) = am^{2/3}n^{1/3} + bm^{5/3}n^{-2/3}, \quad (7)$$

the adjacent branches  $F_n$  and  $F_{n+1}$  cross exactly at

$$m_n = \frac{a}{b} \frac{n^{2/3}(n+1)^{2/3}}{n^{1/3} + (n+1)^{1/3}} = 10\pi \frac{n^{2/3}(n+1)^{2/3}}{n^{1/3} + (n+1)^{1/3}}. \quad (8)$$

The fraction in (8) is strictly increasing in  $n$  (the function  $x^2y^2/(x+y)$  is strictly increasing in each positive variable). Comparing adjacent branches in (7) therefore gives the complete phase diagram: with  $m_0 = 0$ ,  $F_n$  is the unique minimum for

$$m_{n-1} < m < m_n,$$

while  $F_n = F_{n+1}$  are the only minima at  $m = m_n$ . The first thresholds are

| $n$   | 1       | 2       | 3       | 4       | 5       | 6        |
|-------|---------|---------|---------|---------|---------|----------|
| $m_n$ | 22.0670 | 38.3888 | 54.3515 | 70.1996 | 85.9964 | 101.7656 |

In particular  $m_1 = 22.06699868\dots$  is exactly the source's proposed  $m^*$ . Also  $m_n \sim 5\pi n$ , so optimal component masses converge to  $5\pi$ .

### 4 Strict mass stability and actual ball unions

The theorem is stable under structure-preserving perturbations. Suppose  $m_{n-1} < m < m_n$  and  $n \geq 2$ . At the equal partition  $s = m/n$ , the constrained Hessian is

$$\frac{1}{2}f''(s) \sum_{i=1}^n h_i^2, \quad \sum_i h_i = 0.$$

It is positive. Indeed, the inflection mass is

$$\frac{a}{5b} = 2\pi,$$

and, writing  $q = ((n-1)/n)^{1/3} \geq 2^{-1/3}$ ,

$$\frac{m_{n-1}}{n} = 10\pi \frac{q^2}{q+1} > 2\pi.$$

Thus every nontrivial sufficiently small mass-preserving redistribution strictly raises the energy. At a transition, each of the two equal branches has the same within-branch stability, while the droplet count is degenerate.

**Corollary 2** (finite unions of balls). *Among all finite unions of pairwise disjoint balls of total volume  $m$ , the infimum of the liquid-drop energy is  $\min_n F_n(m)$ . It is approached by the optimal number of equal balls whose mutual distances tend to infinity. For  $m \leq m_1$  a single ball attains the restricted infimum. For  $m > m_1$  the restricted infimum is not attained at finite separation.*

*Proof.* For disjoint balls, perimeter is additive (sphere intersections have zero surface measure), while every cross-Coulomb term is nonnegative and is strictly positive for two nonempty components at finite distance. Hence

$$\mathcal{E} \left( \bigcup_{i=1}^N B_i \right) \geq \sum_{i=1}^N f(|B_i|) \geq \min_n F_n(m).$$

Conversely, translate an optimal equal family so that all mutual distances tend to infinity. Every cross term tends to zero, proving equality of the infima and the attainment claims.  $\square$

## 5 Why this does not yet prove the full conjecture

The missing implication is global rather than combinatorial. For an arbitrary component  $\Omega$  of mass  $s$ , perimeter favors the ball but the repulsive Coulomb term favors spreading; one does not know throughout the necessary range that

$$P(\Omega) + D(\Omega) \geq f(s).$$

Replacing each component by a ball is therefore not an energy-decreasing operation. Local shape stability near a ball cannot exclude distant nonspherical competitors.

Frank and Nam (arXiv:2101.02163) proved existence of minimizers throughout  $0 < m \leq m^*$  and proved ball uniqueness there conditional on sharp nonexistence for  $m > m^*$ . Their theorem page is reproduced next. This pinpoints the obstruction: an arbitrary-shape upgrade of the present result would solve the still-open global liquid-drop step, not merely strengthen a partition lemma.

[2], Bonacini and Cristoferi extended (c) and (d) to all  $N \geq 2$ . In [9], Figalli, Fusco, Maggi, Millot and Morini extended (d) to all  $N \geq 2$  and  $\lambda \in (0, N)$ .

Our first new result concerns the existence in (a). Except when  $\lambda > 0$  is small, the existence of minimizers for  $E(m)$  is known only for small  $m$ . In this paper, we extend the existence to what is conjectured to be the optimal range of parameters.

**Theorem 1.** *Let  $N \geq 2$  and  $\lambda \in (0, N)$ . Then the variational problem  $E(m)$  has a minimizer for every  $0 < m \leq m_*$ , where  $m_*$  is defined in (1).*

We will prove Theorem 1 by establishing the strict binding inequality [12]

$$E(m) < E(m_1) + E(m - m_1), \quad \forall 0 < m_1 < m \quad (2)$$

for all  $m < m_*$ . As a by product of our proof, we obtain the following conditional uniqueness of minimizers.

**Theorem 2.** *Let  $N \geq 2$  and  $\lambda \in (0, N)$ . If  $E(m)$  has no minimizer when  $m > m_*$ , then balls are minimizers for  $E(m)$  when  $m \leq m_*$  and they are unique minimizers when  $m < m_*$ .*

So far, the non-existence result in the sharp range  $m > m_*$  is only available for  $\lambda > 0$  small [18, 2]. For larger  $\lambda$  and a nonexplicit range of  $m$ , we have

**Theorem 3.** *Let  $N \geq 2$  and  $\lambda \in (0, N)$  and  $\lambda \leq 2$ . Then there exists a constant  $m_{c_2} \geq m_*$  such that  $E(m)$  does not have a minimizer for all  $m > m_{c_2}$ .*

This result is due to [18, 19, 22] for  $\lambda < 2$  and seems to be unpublished for  $\lambda = 2$ . We will combine the methods in [11] and [18, 19]. It is an open problem whether the nonexistence result also holds for  $2 < \lambda < N$  when  $N \geq 3$ .

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## 2. EXISTENCE

In this section we prove Theorem 1. We will deduce Theorem 1 from the following strict binding inequality.

**Theorem 4.** *Let  $N \geq 2$  and  $\lambda \in (0, N)$ . Then for every  $0 < m < m_*$  with  $m_*$  in (1), we have*

$$E(m) < E(m_1) + E(m - m_1), \quad \forall 0 < m_1 < m. \quad (3)$$

Thanks to [12, Theorem 3.1], the strict binding inequality (3) is a sufficient condition for the existence of minimizers of  $E(m)$ . Moreover, by [12, Theorem 3.4], the set  $\{m > 0 : E(m) \text{ has a minimizer}\}$  is closed in  $(0, \infty)$ . Hence, Theorem 4 implies the existence of minimizers of  $E(m)$  for all  $0 < m \leq m_*$ . Note that the proofs of Theorems 3.1 and 3.4 in [12] extend, without modifications, to the case  $\lambda \neq 1$ ; see Remark 3.7 in that paper.

We will prove the strict binding inequality using a scaling argument, based on the following key observation which uses only the isoperimetric inequality.

## 6 Checks, upgrade log, and scope

**Independent checks.** The companion script verifies  $a/b = 10\pi$ , the first 79 transition identities and their strict ordering, the curvature ratio (6) on a logarithmic grid, and 16,000 random finite partitions against the exact envelope. These computations are diagnostic only; the proof above is analytic.

**Upgrade log.** The investigation made six focused passes: (1) a numerical search over one-small/many-large partitions; (2) the exact negative second-variation argument excluding every mixed critical point; (3) the uniform small-tail merge extending the theorem to countable partitions; (4) the closed-form transition diagram and strict mass-redistribution stability; (5) the positive-cross-interaction extension to actual finite unions of balls; and (6) an arbitrary-shape upgrade attempt, which reduces to the known open global ball-optimality/sharp-nonexistence problem.

**Literature boundary and novelty.** Bounded primary-source searches through 11 August 2026 found the source, Frank–Nam’s 2021 existence and conditional-uniqueness result, prior large-mass nonexistence bounds, and a 2026 improvement of that bound. No source located in the search stated the finite/countable equal-fragmentation theorem with its complete phase diagram. The argument is elementary enough that it may be implicit or folklore; the packet should therefore be treated as a candidate substantial partial result, not a priority claim.

**Human review.** Check the relative-face minimization step, the mixed-point curvature ratio, and the small-tail merge. Also verify that the intended restricted class in Corollary 2 consists of disjoint spherical components; no claim is made for general shapes.

## References

- [1] R. Choksi and M. A. Peletier, *Small volume fraction limit of the diblock copolymer problem II: diffuse-interface functional*, arXiv:1004.4271.
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- [3] R. L. Frank, R. Killip, and P. T. Nam, *Nonexistence of large nuclei in the liquid drop model*, arXiv:1604.03231.